

Vladislav Trifonov

C++/Qt Developer | SCADA, HMI, Industrial Automation

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PROFILE

C++/Qt developer building industrial software: SCADA, HMI and real-time systems. The part of a SCADA system the operator actually looks at is what I do best: mimic diagrams, element editors, alarm handling. Along with the part that has to keep up with the plant behind it: sensor pipelines, threading, latency.

EXPERIENCE

C Developer | JSC "VAZSYSTEM"

Feb 2026 - Present · Samara

"Kaskad" SCADA system

- Building operator panels for WinCC OA in the XML panel format and writing the CTRL scripts that drive them
- Writing WinCC OA modules and C++/Qt shared libraries (DLL/.so) for the system core
- Designed the PostgreSQL schema and introduced Flyway
- Set up the project's CI pipeline
- Reviewing code in a team of 4 developers; maintaining the project's architecture and design documentation

Stack: C++, Qt, WinCC OA (XML panels, CTRL/Ctrl++), OPC UA, PostgreSQL, Flyway, CMake, Windows/Linux, CI/CD

C++/Qt Developer (SCADA) | RASU

Jun 2024 - Jan 2026 · Moscow

A SCADA platform for industrial facilities

- Built the diagnostics module from scratch with one other developer, as a separate application: health of the field sensors and of the links to the controllers, presented as state tables with signal tracing on top. From an empty repository to a working version in three months
- Owned the graphical mimic diagram editor and extended it with new modules. Operators compose schematics from 100+ elements, with several diagrams open and updating live at the same time
- Built the alarm handling module: under a second from the incoming event to the object changing state on the mimic diagram
- Wrote the gRPC module for live sensor signal subscriptions
- Replaced the per-widget change handlers (QLineEdit::textChanged, QComboBox::currentIndexChanged, QSpinBox::valueChanged) with one templated debounced handler. Every keystroke used to fire its own server request and flood the event loop; now a single request goes out once the user stops typing
- Tracked down and fixed memory leaks with Valgrind
- Designed module architecture before implementation

Stack: C++11, Qt 5.12+ (Widgets, QML, Model/View, multithreading), gRPC, MQTT, CMake, Windows/Linux, Valgrind

C++/Qt Developer | LADUGA

May 2023 - Jun 2024 · Samara

Desktop software for the automotive industry. First job, combined with university

- Built the plugin system: third-party extensions load without touching the core
- Wrote the extension management UI: install, remove, configure
- Built Qt Model/View components: custom ComboBox and TreeView, delegates, proxy models
- Diagnosed and fixed memory leaks with the Visual Studio diagnostic tools

Stack: C++11, Qt 5 (Widgets, Model/View), MSVC, Visual Studio, CMake, Windows, Git

SKILLS

Languages, frameworks: C++11/17, Qt 5/6 (Widgets, QML/Qt Quick, Model/View, multithreading), CTRL/Ctrl++, OOP, design patterns

Industrial: SCADA, HMI, WinCC OA, OPC UA, Modbus, real-time systems, mimic diagrams, process control

Data, protocols: gRPC, Protobuf, MQTT, TCP/IP, PostgreSQL, Flyway

Build, tooling: CMake, MSVC/Visual Studio, Git, Docker, CI/CD, QtTest, clang-tidy, Valgrind, Linux, Windows

EDUCATION

Samara State Technical University - Bachelor's degree, Software Engineering, Sep 2021 - Jun 2025